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**Stochastic Volatility Models Based on Martingale Optimal
Transport**

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1 Introduction

The construction of arbitrage-free asset price models calibrated to observed option prices is a central problem in financial mathematics. Under a risk-neutral measure, the discounted price process must be a martingale, while European option prices determine the one-dimensional marginal laws of the underlying asset at traded maturities. Thus, the calibration problem can be viewed as the problem of constructing a martingale with prescribed marginal distributions.

One of the main industrial tools for this problem is the Dupire local volatility model [21]. It describes the asset price by the Markovian diffusion

$$dS_t = rS_t dt + \sigma_{\text{loc}}(t, S_t)S_t dW_t,$$

where the local volatility function is recovered from the call price surface $C(T, K)$ by Dupire’s formula

$$\sigma_{\text{loc}}^2(T, K) = \frac{\partial_T C(T, K) + rK \partial_K C(T, K)}{\frac{1}{2}K^2 \partial_{KK} C(T, K)}.$$

The model calibrates exactly to vanilla option prices, but it has two standard drawbacks. First, it requires an arbitrage-free interpolation and extrapolation of option prices. Second, the local volatility model produces unrealistic smile dynamics.

A classical way to address the second issue is to combine local volatility with a stochastic volatility factor. This leads to local stochastic volatility models, for example from [35, 29, 14]. These models enrich the dynamics, but they still rely on an interpolated implied volatility surface and require additional calibration steps, in particular the evaluation of conditional expectations [38].

A different approach is given by the Bass local volatility model [17]. Instead of interpolating the implied volatility surface in time, the Bass construction uses the option-implied marginal laws at the traded maturities. From a practical point of view, this gives a local volatility model with a stable and fast calibration procedure [1, 47]. From a theoretical point of view, the resulting process appears as a solution to the dynamic optimal transport problem known as the Martingale Benamou–Brenier problem, which can be interpreted as minimizing a quadratic divergence from Brownian motion.

The present paper follows the Martingale Benamou–Brenier perspective, but replaces the Brownian reference dynamics with a more general martingale reference. In particular, we focus on reference martingales generated by stochastic volatility models. The aim is to combine the marginal calibration mechanism of Bass local volatility with richer volatility dynamics.

The paper is organized as follows. Section 1 reviews existing applications of optimal transport methods in financial mathematics. Section 2 formulates the generalized martingale Benamou–Brenier problem with a stochastic volatility reference process and discusses the one-marginal case as an illustration. Section 3 derives the static weak optimal transport formulation, proves its equivalence with the dynamic problem, and establishes the existence of static optimizers. Section 4 studies Bass-type optimizers and proves the corresponding verification theorem. Section 5 presents a practical calibration procedure based on a fixed-point problem. Finally, Section 6 provides numerical examples, including calibrations to SABR, Heston, and SVI-implied marginals.

1.1 Related Literature

Optimal transport theory dates back to the works [44, 41], with a central role in its modern formulation played by [16, 13, 23, 43]. For general references, see [15, 52, 53, 46]. The transportation problem with a martingale constraint was first addressed by Strassen’s theorem [48], which characterizes the existence of martingale couplings in terms of the convex order of measures. The systematic development of martingale optimal transport was initiated in [12]. The martingale constraint arises naturally in mathematical finance, where it has been widely applied to model-free option pricing, for example, [36, 37, 9, 22, 20, 34, 31].

The Martingale Benamou–Brenier problem is the martingale analogue of the classical Benamou–Brenier formulation of optimal transport. It was introduced in [3, 39] and provides a dynamic formulation of martingale optimal transport. Its optimizer is closely related to the Bass martingale, which first appeared as a solution to the Skorokhod embedding problem from [8]. The structure, duality, and stability of the Martingale Benamou–Brenier problem have been studied in [4, 7, 6]. The role of irreducibility and the decomposition of martingale transport problems were studied in [12, 19, 45].

The Bass local volatility model was introduced in [17] as a relaxation of the Dupire local volatility model [21]. Its calibration is reduced to a fixed-point problem studied in detail in [1], while numerical and implementation aspects are considered in [47]. Further developments of the Bass construction and related martingale transport methods appear in [40, 5, 51, 30].

A more general dynamic transport problem is formulated within semimartingale optimal transport: one seeks a process that satisfies marginal constraints while optimizing a functional depending on the characteristics of the process. This viewpoint was introduced in [49] and developed in the context of volatility models in [27, 25, 28, 26]. Related reference-model approaches also arise in Schrödinger bridge and entropic optimal transport problems [32, 42, 18]. Recent works connecting Schrödinger bridges and Bass-type martingale transport include [33, 2, 50].

1.2 Martingale Benamou–Brenier problem

We formulate an extension of the one-dimensional Martingale Benamou–Brenier problem, which will later be used to construct our model on a single time interval between two maturities. To make the presentation self-contained and to fix notation, we first recall the basic definitions and results underlying the classical Martingale Benamou–Brenier problem.

Let $\mathcal{P}_p(\mathbb{R}^d)$ denote the set of probability measures on $\mathbb{R}^d$ with finite p -th moment.

Definition 1.1 (Convex order). Let $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$. We write $\mu \preceq_{cx} \nu$ if

$$\int_{\mathbb{R}^d} \varphi(x) \mu(dx) \leq \int_{\mathbb{R}^d} \varphi(x) \nu(dx)$$

for every convex function $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ for which both integrals are well defined.

Denote by $\text{Cpl}(\mu, \nu)$ the set of all couplings of μ and ν , i.e., the set of probability measures π on $\mathbb{R}^d \times \mathbb{R}^d$ such that $\pi(A \times \mathbb{R}^d) = \mu(A)$ and $\pi(\mathbb{R}^d \times B) = \nu(B)$ for all measurable $A, B \subseteq \mathbb{R}^d$.

Theorem 1.2 (Strassen’s theorem). *Let $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$. There exists a martingale coupling $\pi \in \text{Cpl}(\mu, \nu)$, that is, a probability measure π on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν such that*

$$\int_{\mathbb{R}^d} y \pi_x(dy) = x \quad \text{for } \mu\text{-a.e. } x,$$

where $\pi(dx, dy) = \mu(dx) \pi_x(dy)$, if and only if $\mu \preceq_{cx} \nu$.

Let now $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$ satisfy $\mu_0 \preceq_{cx} \mu_1$. We recall the dynamic formulation of the Martingale Benamou–Brenier problem from [3]. Let

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0,1]}, \mathbb{P})$$

be a filtered probability space supporting a Brownian motion B . We assume that $\mathcal{F}_0$ is rich enough to support an atomless random variable, so that any initial law $\mu_0 \in \mathcal{P}_2(\mathbb{R})$ can be realized by an $\mathcal{F}_0$ -measurable random variable X_0 . We also assume that B is independent of $\mathcal{F}_0$.

Define the admissible class

$$\mathcal{X}(\mu_0, \mu_1) = \left\{ X : X_t = X_0 + \int_0^t \sigma_s dB_s, \quad X_0 \sim \mu_0, \quad X_1 \sim \mu_1, \quad \mathbb{E} \int_0^1 \sigma_s^2 ds < \infty \right\},$$

where σ is progressively measurable with respect to $(\mathcal{F}_t)$.

Definition 1.3 (Martingale Benamou–Brenier problem). The Martingale Benamou–Brenier problem is

$$\inf_{X \in \mathcal{X}(\mu_0, \mu_1)} \mathbb{E} \left[\int_0^1 (\sigma_s - 1)^2 ds \right]. \quad (\text{MBB-min})$$

This problem selects, among all martingales with prescribed initial and terminal laws, the one whose volatility is closest to the Brownian volatility. Equation (MBB-min) has an important interpretation in terms of the angle bracket of the martingale, so we recall the definition.

Definition 1.4 (Angle bracket). Let $M = (M_t)_{t \in [0,1]}$ be a square-integrable martingale. Its angle bracket $\langle M \rangle = (\langle M \rangle_t)_{t \in [0,1]}$ is the unique predictable, increasing process starting from 0 such that

$$M_t^2 - \langle M \rangle_t$$

is a martingale.

Proposition 1.5. *The Martingale Benamou–Brenier problem admits an equivalent formulation in terms of the angle bracket:*

$$\inf_{X \in \mathcal{X}(\mu_0, \mu_1)} \mathbb{E} \left[\int_0^1 (\sigma_s - 1)^2 ds \right] = \inf_{X \in \mathcal{X}(\mu_0, \mu_1)} \mathbb{E} \langle X - B \rangle_1,$$

and also an equivalent maximization form:

$$\sup_{X \in \mathcal{X}(\mu_0, \mu_1)} \mathbb{E} \left[\int_0^1 \sigma_s ds \right] = \sup_{X \in \mathcal{X}(\mu_0, \mu_1)} \mathbb{E} \langle X, B \rangle_1. \quad (\text{MBB})$$

Proof. Expanding the square gives

$$\mathbb{E} \left[\int_0^1 (\sigma_s - 1)^2 ds \right] = \mathbb{E} \left[\int_0^1 \sigma_s^2 ds \right] + 1 - 2 \mathbb{E} \left[\int_0^1 \sigma_s ds \right].$$

Since $\mathbb{E} \int_0^1 \sigma_s^2 ds = \mathbb{E} \langle X \rangle_1 = \mathbb{E}[X_1^2] - \mathbb{E}[X_0^2]$ is fixed by the marginals, minimizing (MBB-min) is equivalent to maximizing $\mathbb{E} \left[\int_0^1 \sigma_s ds \right]$. From the dynamics of X , we have $\langle X - B \rangle_t = \int_0^t (\sigma_s - 1)^2 ds$, so

$$\mathbb{E} \langle X - B \rangle_1 = \mathbb{E} \left[\int_0^1 (\sigma_s - 1)^2 ds \right].$$

The maximization form in terms of $\langle X, B \rangle$ then follows by polarization. $\square$

This bracket formulation motivates our extension. In the classical problem, the reference martingale is Brownian motion B . In the generalized problem below, we replace this Brownian reference by a fixed continuous square-integrable martingale Y .

2 Statement of the problem

We now formulate an extension of the Martingale Benamou–Brenier problem. The classical problem can be interpreted as the minimization of the quadratic-variation distance to Brownian motion. We keep this interpretation, but replace the Brownian reference by a fixed continuous square-integrable martingale.

Generalized Martingale Benamou–Brenier problem Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$ satisfy $\mu_0 \preceq_{cx} \mu_1$. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0,1]}, \mathbb{P})$ be a fixed filtered probability space, and let $Y = (Y_t)_{t \in [0,1]}$ be a continuous square-integrable $(\mathcal{F}_t)$ -martingale. We consider the class

$$\mathcal{X}^Y(\mu_0, \mu_1) = \left\{ X : \begin{array}{l} X \text{ is a continuous square-integrable } (\mathcal{F}_t)\text{-martingale,} \\ X_0 \sim \mu_0, \quad X_1 \sim \mu_1 \end{array} \right\}.$$

The superscript Y emphasizes that the admissible martingales are defined on the same filtered probability space as the reference martingale Y .

Definition 2.1 (Generalized Martingale Benamou–Brenier problem). The generalized Martingale Benamou–Brenier problem with reference martingale Y is

$$V_{\text{MBB}}^Y(\mu_0, \mu_1) = \inf_{X \in \mathcal{X}^Y(\mu_0, \mu_1)} \mathbb{E} \langle X - Y \rangle_1. \quad (\text{GMBB})$$

The bracket $\langle X - Y \rangle$ is well defined because both processes are constructed on the same filtered probability space. In the static formulation below, the relevant object will be the coupling between the terminal value of X and the increment $Y_1 - Y_0$ of the reference process.

Stochastic volatility Martingale Benamou–Brenier problem We now specify the generalized problem in the case where a stochastic volatility model generates the reference martingale. Again, let

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0,1]}, \mathbb{P})$$

be a filtered probability space such that $\mathcal{F}_0$ is rich enough to support the required initial random variables. Let B and W be Brownian motions with correlation coefficient ρ , independent of $\mathcal{F}_0$. We work with the filtration

$$\mathcal{F}_t = \sigma(\mathcal{F}_0, B_s, W_s : 0 \leq s \leq t).$$

Let ξ_0 be an $\mathcal{F}_0$ -measurable random variable with law β_0 . We define the stochastic volatility process by

$$d\xi_t = b(\xi_t) dt + \eta(\xi_t) dW_t,$$

and assume that the coefficients are such that this equation has a unique strong solution and $\mathbb{E} \int_0^1 \xi_t^2 dt < \infty$. The reference martingale is defined by

$$Y_t = \int_0^t \xi_s dB_s.$$

In particular, $Y_0 = 0$, Y is a continuous square-integrable martingale, and $\langle Y \rangle_t = \int_0^t \xi_s^2 ds$.

Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$ satisfy $\mu_0 \preceq_{cx} \mu_1$. We fix a coupling $\kappa \in \text{Cpl}(\mu_0, \mu_1)$ and choose an $\mathcal{F}_0$ -measurable pair (X_0, ξ_0) such that $\text{Law}(X_0, \xi_0) = \kappa$. This is possible by the richness assumption on $\mathcal{F}_0$.

We denote by $\mathcal{X}(\mu_0, \mu_1; \kappa)$ the class of all martingales:

$$\mathcal{X}(\mu_0, \mu_1; \kappa) = \left\{ X : \begin{array}{l} X_t = X_0 + \int_0^t \sigma_s^{(1)} dB_s + \int_0^t \sigma_s^{(2)} dW_s, \\ \text{Law}(X_0, \xi_0) = \kappa, \quad X_1 \sim \mu_1, \quad \mathbb{E} \int_0^1 (\sigma_s^{(1)})^2 + (\sigma_s^{(2)})^2 ds < \infty \end{array} \right\},$$

where $\sigma_s^{(1)}$ and $\sigma_s^{(2)}$ are $(\mathcal{F}_t)$ -predictable processes. The condition $X_0 \sim \mu_0$ is contained in $\text{Law}(X_0, \xi_0) = \kappa$.

Definition 2.2 (Stochastic volatility MBB problem). The stochastic volatility Martingale Benamou–Brenier problem with initial coupling κ is

$$\inf_{X \in \mathcal{X}(\mu_0, \mu_1; \kappa)} \mathbb{E} \langle X - Y \rangle_1. \quad (1)$$

Equivalently, we use the maximization form

$$V_{\text{MBB}}^\kappa(\mu_0, \mu_1) = \sup_{X \in \mathcal{X}(\mu_0, \mu_1; \kappa)} \mathbb{E} \langle X, Y \rangle_1. \quad (\text{SV-MBB})$$

Since

$$dX_t = \sigma_t^{(1)} dB_t + \sigma_t^{(2)} dW_t, \quad dY_t = \xi_t dB_t,$$

the cost becomes

$$\mathbb{E} \langle X - Y \rangle_1 = \mathbb{E} \int_0^1 \left((\sigma_s^{(1)} - \xi_s)^2 + 2\rho(\sigma_s^{(1)} - \xi_s)\sigma_s^{(2)} + (\sigma_s^{(2)})^2 \right) ds. \quad (2)$$

Using the orthogonal representation of the Brownian motions,

$$W_t = \rho B_t + \sqrt{1 - \rho^2} Z_t, \quad \rho \in (-1, 1),$$

where B and Z are independent Brownian motions, we obtain a cleaner form. Indeed, rewriting the integrand in terms of

$$\alpha_s = \sigma_s^{(1)} + \rho \sigma_s^{(2)}, \quad \gamma_s = \sqrt{1 - \rho^2} \sigma_s^{(2)},$$

the dynamics become

$$dX_t = \alpha_t dB_t + \gamma_t dZ_t, \quad dY_t = \xi_t dB_t,$$

and the cost simplifies to

$$\mathbb{E} \langle X - Y \rangle_1 = \mathbb{E} \int_0^1 ((\alpha_s - \xi_s)^2 + \gamma_s^2) ds. \quad (3)$$

Thus the problem selects, among all admissible martingales with prescribed endpoint marginals and fixed initial coupling κ , the one whose volatility vector (α_s, γ_s) is closest to the reference vector $(\xi_s, 0)$ in the mean-square sense.

The reference martingale starts at $Y_0 = 0$, initial randomness is instead carried by the pair (X_0, ξ_0) through the coupling $\kappa = \text{Law}(X_0, \xi_0)$, which will play a key role in the static formulation below.

Remark 2.3 (Choice of the initial coupling). The coupling κ is part of the model specification. Suppose the reference stochastic volatility model is already accurately calibrated at the initial time t , which is taken as the starting point of the calibration. The reference model $(\tilde{Y}_t, \tilde{\xi}_t)$ has a fixed joint distribution at time t , with marginal $\tilde{Y}_t \sim \nu$. Assume that the Kantorovich–Wasserstein distance $\mathcal{W}_2(\mu_0, \nu)$ satisfies $\mathcal{W}_2(\mu_0, \nu) < \varepsilon$, a quantity that reflects the calibration accuracy. Let f denote the Brenier map from ν to μ_0 , and set

$$\kappa = \text{Law}(f(\tilde{Y}_t), \tilde{\xi}_t)$$

to preserve the dependence between the reference price and the volatility state.

2.1 Example: One marginal

For illustration, we first consider the simple case where the initial law is a Dirac measure, $\mu_0 = \delta_0$. In this setting, the construction of the optimizer is analogous to the classical Bass martingale. Different choices of the reference martingale lead to different Bass-type martingales.

Let F_ξ denote the distribution function of a random variable ξ .

Example 2.4 (Classical Bass martingale). Let the reference martingale be a standard Brownian motion

$$Y_t = W_t = \int_0^t dW_s, \quad t \in [0, 1].$$

Let $\mu_1 \in \mathcal{P}_2(\mathbb{R})$ be centred, and define the monotone transport map

$$\phi(x) = F_{\mu_1}^{-1} \circ F_{W_1}(x).$$

Then ϕ pushes $\text{Law}(W_1)$ forward to μ_1 . The classical Bass martingale is defined by

$$M_t = \mathbb{E}[\phi(W_1) \mid \mathcal{F}_t^W], \quad M_0 = \mathbb{E}[\phi(W_1)] = 0, \quad M_1 = \phi(W_1) \sim \mu_1,$$

where $\mathcal{F}_t^W$ is the filtration generated by the Brownian motion W_t . If

$$u(t, x) = \mathbb{E}[\phi(W_1) \mid W_t = x],$$

then $M_t = u(t, W_t)$. By Itô's formula,

$$M_t = M_0 + \int_0^t \partial_x u(s, W_s) dW_s.$$

Example 2.5 (Bass martingale associated with a reference martingale Y). Let the reference martingale be the arithmetic stochastic volatility model

$$Y_t = \int_0^t \xi_s dB_s, \quad t \in [0, 1],$$

where ξ is progressively measurable. Let $\mu_1 \in \mathcal{P}_2(\mathbb{R})$ be centred, and define

$$\phi(x) = F_{\mu_1}^{-1} \circ F_{Y_1}(x).$$

Then ϕ is the monotone transport map pushing $\text{Law}(Y_1)$ forward to μ_1 . The Bass martingale associated with the reference process Y is defined by

$$M_t = \mathbb{E}[\phi(Y_1) \mid \mathcal{F}_t], \quad M_0 = \mathbb{E}[\phi(Y_1)] = 0, \quad M_1 = \phi(Y_1) \sim \mu_1.$$

If the conditional expectation admits a Markovian representation

$$M_t = v(t, Y_t, \xi_t), \quad v(t, y, a) = \mathbb{E}[\phi(Y_1) \mid Y_t = y, \xi_t = a],$$

and if ξ has dynamics

$$d\xi_t = b(t, Y_t, \xi_t) dt + \eta(t, Y_t, \xi_t) dZ_t,$$

where Z is a Brownian motion, possibly correlated with B , then Itô's formula gives

$$M_t = M_0 + \int_0^t \partial_y v(s, Y_s, \xi_s) \xi_s dB_s + \int_0^t \partial_a v(s, Y_s, \xi_s) \eta(s, Y_s, \xi_s) dZ_s.$$

Thus we obtain a Bass-type construction associated with the reference martingale Y .

Proposition 2.6. *Let $\mu \in \mathcal{P}_2(\mathbb{R})$ be centred. For a given reference martingale Y , assume that $\text{Law}(Y_1)$ is atomless, and let*

$$\phi(x) = F_\mu^{-1} \circ F_{Y_1}(x)$$

be the monotone transport map from $\text{Law}(Y_1)$ to μ . Then the process

$$M_t = \mathbb{E}[\phi(Y_1) \mid \mathcal{F}_t]$$

is a solution of the one-marginal generalized MBB problem

$$V_{\text{MBB}}^Y(\delta_0, \mu) = \inf_{X \in \mathcal{X}^Y(\delta_0, \mu)} \mathbb{E}\langle X - Y \rangle_1.$$

Proof. Let $X \in \mathcal{X}^Y(\delta_0, \mu)$. Since $X_0 = 0$, $Y_0 = 0$, and X and Y are continuous square-integrable martingales, we have

$$\mathbb{E}\langle X - Y \rangle_1 = \mathbb{E}[(X_1 - Y_1)^2] - \mathbb{E}[(X_0 - Y_0)^2] = \mathbb{E}[(X_1 - Y_1)^2].$$

Therefore

$$\inf_{X \in \mathcal{X}^Y(\delta_0, \mu)} \mathbb{E}\langle X - Y \rangle_1 = \inf_{X \in \mathcal{X}^Y(\delta_0, \mu)} \mathbb{E}[(X_1 - Y_1)^2].$$

The right-hand side depends only on the joint law of (X_1, Y_1) and is bounded below by the quadratic optimal transport problem

$$\inf_{\pi \in \text{Cpl}(\text{Law}(Y_1), \mu)} \int_{\mathbb{R} \times \mathbb{R}} |x - y|^2 \pi(dx, dy).$$

In one dimension, since $\text{Law}(Y_1)$ is atomless, the unique optimal coupling for the quadratic cost is induced by the monotone transport map

$$\phi(y) = F_\mu^{-1} \circ F_{Y_1}(y).$$

Hence the lower bound is attained by choosing

$$M_1 = \phi(Y_1) \sim \mu.$$

Now define

$$M_t = \mathbb{E}[M_1 \mid \mathcal{F}_t] = \mathbb{E}[\phi(Y_1) \mid \mathcal{F}_t].$$

Then M is a square-integrable martingale and satisfies

$$M_0 = \mathbb{E}[M_1] = \int_{\mathbb{R}} x \mu(dx) = 0, \quad M_1 \sim \mu.$$

Thus $M \in \mathcal{X}^Y(\delta_0, \mu)$ and attains the lower bound above. Therefore M solves $V_{\text{MBB}}^Y(\delta_0, \mu)$. $\square$

3 Two marginals

In this section we derive a static formulation of the generalized Martingale Benamou–Brenier problem. We then prove its equivalence with the dynamic formulation and establish existence of optimizers. For the classical problem (MBB), the authors of [3] use a reformulation in terms of weak optimal transport. We show that a similar reformulation is available in the present setting for (SV-MBB).

3.1 Weak optimal transport

For later use, we first recall the weak optimal transport problem, introduced in [24] and further developed in [11, 10].

Let X, Y be Polish spaces and fix $\mu \in \mathcal{P}_p(X)$. For $\pi \in \text{Cpl}(\mu, \nu)$, write its disintegration as

$$\pi(dx, dy) = \mu(dx) \pi_x(dy).$$

Given a cost function

$$C : X \times \mathcal{P}_p(Y) \rightarrow \mathbb{R},$$

the corresponding weak optimal transport problem is to find a coupling π^* that solves

$$\inf_{\pi \in \text{Cpl}(\mu, \nu)} \int_X C(x, \pi_x) \mu(dx).$$

The existence and stability of this problem rely on lower semicontinuity of the cost C , together with convexity in its second argument.

Theorem 3.1 (Lower semicontinuity of WOT, [11]). *Assume that C is bounded from below, lower semicontinuous on $X \times \mathcal{P}_p(Y)$, and convex in the second variable. Fix $\mu \in \mathcal{P}_p(X)$. If $\pi_n \rightarrow \pi$ in $\mathcal{P}_p(X \times Y)$ and $(\text{proj}_X)_\# \pi_n = \mu$ for all n , then*

$$\int_X C(x, \pi_x) \mu(dx) \leq \liminf_{n \rightarrow \infty} \int_X C(x, \pi_{n,x}) \mu(dx).$$

In (MBB), the optimization is restricted to $\pi \in \text{Cpl}_M(\mu, \nu)$, where $\text{Cpl}_M(\mu, \nu)$ denotes the set of martingale couplings between μ and ν . Therefore, the relevant static problem is a weak martingale optimal transport problem.

3.2 Static problem

The classical Martingale Benamou–Brenier problem admits a static weak optimal transport formulation. In the one-dimensional case, if γ denotes the law of the Brownian increment $B_1 - B_0$, then the static value is

$$V_{WOT}(\mu_0, \mu_1) = \sup_{\pi \in \text{Cpl}_M(\mu_0, \mu_1)} \int \sup_{q \in \text{Cpl}(\pi_x, \gamma)} \int yz q(dy, dz) \mu_0(dx), \quad (\text{WOT})$$

where $\pi(dx, dy) = \mu_0(dx)\pi_x(dy)$. It was proved in [3] that (WOT) is equivalent to the dynamic maximization problem (MBB). The static problem chooses the martingale coupling whose conditional terminal kernel has maximal covariance with the reference increment.

In the present setting, the Brownian reference is replaced by the stochastic volatility martingale

$$Y_t = Y_0 + \int_0^t \xi_s dB_s.$$

Therefore, the Brownian increment law γ is replaced by the conditional law of the increment

$$Z = Y_1 - Y_0 = \int_0^1 \xi_t dB_t.$$

Unlike in the classical case, this law is not fixed: it depends on the initial volatility state ξ_0 . Thus we define the increment law θ_a conditional on the initial volatility value $\xi_0 = a$. For each $a \in \text{supp}(\beta_0)$, let ξ^a solve

$$d\xi_t^a = b(\xi_t^a) dt + \eta(\xi_t^a) dW_t, \quad \xi_0^a = a,$$

and define

$$\theta_a = \text{Law}(Y_1 - Y_0 \mid \xi_0 = a) = \text{Law}\left(\int_0^1 \xi_t^a dB_t\right). \quad (4)$$

Throughout this section we assume that, conditionally on $\xi_0 = a$, the law of $Y_1 - Y_0$ does not depend on X_0 . This is the case when the initial filtration $\mathcal{F}_0$ is independent of the Brownian increments driving the reference process.

We now describe the admissible static variables. Let $\lambda \in \mathcal{P}(\mathbb{R}^3)$ be the joint law of

$$(X_0, \xi_0, X_1).$$

We require the joint law of the initial variables (X_0, ξ_0) to be the prescribed coupling κ , and the terminal law of X_1 to be μ_1 . Writing the disintegration of λ with respect to its first two coordinates as

$$\lambda(dx, da, dy) = \kappa(dx, da) \lambda_{x,a}(dy),$$

the measure $\lambda_{x,a}$ is the conditional law of X_1 given $(X_0, \xi_0) = (x, a)$.

Definition 3.2 (Static admissible set). Define

$$\text{Cpl}_{M_1}(\kappa, \mu_1) = \left\{ \lambda \in \mathcal{P}(\mathbb{R}^3) : \lambda \in \text{Cpl}(\kappa, \mu_1), \int_{\mathbb{R}} y \lambda_{x,a}(dy) = x \text{ } \kappa\text{-a.e.} \right\},$$

where $\lambda(dx, da, dy) = \lambda_{x,a}(dy)\kappa(dx, da)$ is the disintegration of λ with respect to its marginal κ . Equivalently, $\text{Cpl}_{M_1}(\kappa, \mu_1)$ is the set of couplings $\lambda \in \text{Cpl}(\kappa, \mu_1)$ satisfying the martingale constraint in the first coordinate.

The last condition is the martingale constraint for the two-step process (X_0, X_1) .

Definition 3.3 (Generalized static weak transport problem). The stochastic volatility weak transport problem is

$$\inf_{\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)} \iint \mathcal{W}_2^2(\lambda_{x,a}, \theta_a) \kappa(dx, da). \quad (5)$$

where $\mathcal{W}_2(\lambda_{x,a}, \theta_a)$ denotes the Kantorovich–Wasserstein distance between the measures $\lambda_{x,a}$ and θ_a .

Proposition 3.4. *The problem (5) admits a covariance maximization form*

$$V_{WOT}^\kappa(\mu_0, \mu_1) = \sup_{\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)} \iint \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a)} \int yz q(dy, dz) \kappa(dx, da). \quad (\text{SV-WOT})$$

Proof. For every admissible $\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)$, the identity

$$W_2^2(\lambda_{x,a}, \theta_a) = \int y^2 \lambda_{x,a}(dy) + \int z^2 \theta_a(dz) - 2 \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a)} \int yz q(dy, dz)$$

holds for κ -a.e. (x, a) . Therefore

$$\begin{aligned} \iint W_2^2(\lambda_{x,a}, \theta_a) \kappa(dx, da) &= \int y^2 \mu_1(dy) + \iint z^2 \theta_a(dz) \kappa(dx, da) \\ &\quad - 2 \iint \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a)} \int yz q(dy, dz) \kappa(dx, da). \end{aligned}$$

The first term is fixed by the terminal marginal μ_1 . The second term is fixed by the reference process, since

$$\iint z^2 \theta_a(dz) \kappa(dx, da) = \mathbb{E}[(Y_1 - Y_0)^2].$$

Hence, minimizing (5) is equivalent to maximizing (SV-WOT). The two problems have the same optimizers, and their values differ by a fixed constant

$$\mathbb{E}[X_1^2] + \mathbb{E}[(Y_1 - Y_0)^2].$$

□

Compared with the classical formulation (WOT), the fixed Gaussian law γ is replaced by the family of reference increment kernels (θ_a) , parametrized by the initial volatility state.

3.3 Equivalence of the dynamic and static problems

We now show that the static problem (SV-WOT) is equivalent to the dynamic maximization of the quadratic covariation.

Theorem 3.5 (Dynamic-static equivalence). *Let $\kappa \in \text{Cpl}(\mu_0, \beta_0)$ be fixed. Assume that (X_0, ξ_0) is independent of the Brownian increments driving the reference process. Then*

$$V_{MBB}^\kappa(\mu_0, \mu_1) = V_{WOT}^\kappa(\mu_0, \mu_1),$$

where

$$V_{MBB}^\kappa(\mu_0, \mu_1) = \sup_{X \in \mathcal{X}(\mu_0, \mu_1; \kappa)} \mathbb{E}\langle X, Y \rangle_1.$$

Proof. We prove the two inequalities separately. First, for every $X \in \mathcal{X}(\mu_0, \mu_1; \kappa)$, since X and Y are continuous square-integrable martingales,

$$\mathbb{E}\langle X, Y \rangle_1 = \mathbb{E}[(X_1 - X_0)(Y_1 - Y_0)].$$

Moreover, X_0 is $\mathcal{F}_0$ -measurable and Y is a martingale, so

$$\mathbb{E}[X_0(Y_1 - Y_0)] = \mathbb{E}[X_0 \mathbb{E}(Y_1 - Y_0 \mid \mathcal{F}_0)] = 0.$$

Therefore

$$\mathbb{E}\langle X, Y \rangle_1 = \mathbb{E}[X_1(Y_1 - Y_0)]. \quad (6)$$

This identity explains why the static formulation couples the conditional terminal law of X_1 with the conditional increment law of the reference martingale.

Step 1. We prove that

$$V_{WOT}^\kappa(\mu_0, \mu_1) \geq V_{MBB}^\kappa(\mu_0, \mu_1).$$

Let $X \in \mathcal{X}(\mu_0, \mu_1; \kappa)$ be arbitrary and define

$$\lambda = \text{Law}(X_0, \xi_0, X_1).$$

Disintegrate λ with respect to $\kappa = \text{Law}(X_0, \xi_0)$ as $\lambda(dx, da, dy) = \kappa(dx, da)\lambda_{x,a}(dy)$. Since X is a martingale and (X_0, ξ_0) is $\mathcal{F}_0$ -measurable,

$$\mathbb{E}[X_1 \mid X_0, \xi_0] = \mathbb{E}[\mathbb{E}[X_1 \mid \mathcal{F}_0] \mid X_0, \xi_0] = X_0.$$

Hence

$$\int y \lambda_{x,a}(dy) = x, \quad \kappa\text{-a.e. } (x, a),$$

and therefore $\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)$.

Set $Z = Y_1 - Y_0$. For κ -a.e. (x, a) , define

$$q_{x,a} = \text{Law}(X_1, Z \mid X_0 = x, \xi_0 = a).$$

The first marginal of $q_{x,a}$ is $\lambda_{x,a}$, and the second marginal is θ_a . Thus

$$q_{x,a} \in \text{Cpl}(\lambda_{x,a}, \theta_a).$$

Consequently,

$$\sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a)} \int yz q(dy, dz) \geq \int yz q_{x,a}(dy, dz) = \mathbb{E}[X_1 Z \mid X_0 = x, \xi_0 = a].$$

Integrating with respect to $\kappa(dx, da)$, we obtain

$$V_{WOT}^\kappa(\mu_0, \mu_1) \geq \mathbb{E}[X_1 Z] = \mathbb{E}[X_1(Y_1 - Y_0)]$$

Since $X \in \mathcal{X}(\mu_0, \mu_1; \kappa)$ was arbitrary,

$$V_{WOT}^\kappa(\mu_0, \mu_1) \geq V_{MBB}^\kappa(\mu_0, \mu_1).$$

Step 2. We prove the reverse inequality:

$$V_{MBB}^\kappa(\mu_0, \mu_1) \geq V_{WOT}^\kappa(\mu_0, \mu_1).$$

Let $\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)$ be arbitrary. For κ -a.e. (x, a) , choose an optimal transport plan $q_{x,a} \in \text{Cpl}(\lambda_{x,a}, \theta_a)$ such that

$$\int yz q_{x,a}(dy, dz) = \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a)} \int yz q(dy, dz).$$

Assume that this family of optimal couplings can be chosen measurably in (x, a) and realized conditionally on (X_0, ξ_0, Z) . That is, assume there exists a random variable X_1^λ such that, for κ -a.e. (x, a) ,

$$\text{Law}(X_1^\lambda, Z \mid X_0 = x, \xi_0 = a) = q_{x,a}.$$

Then

$$\text{Law}(X_1^\lambda \mid X_0 = x, \xi_0 = a) = \lambda_{x,a},$$

and consequently $\text{Law}(X_1^\lambda) = \mu_1$. Moreover, since $\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)$,

$$\mathbb{E}[X_1^\lambda \mid X_0 = x, \xi_0 = a] = \int y \lambda_{x,a}(dy) = x.$$

Hence $\mathbb{E}[X_1^\lambda \mid X_0, \xi_0] = X_0$.

Define the process:

$$X_t^\lambda = \mathbb{E} \left[X_1^\lambda \mid \mathcal{F}_t^{B,W} \vee \sigma(X_0, \xi_0) \right], \quad 0 \leq t \leq 1.$$

Then X^λ is a square-integrable martingale with $X_0^\lambda = \mathbb{E}[X_1^\lambda \mid X_0, \xi_0] = X_0$. Therefore

$$\text{Law}(X_0^\lambda, \xi_0) = \kappa, \quad \text{Law}(X_1^\lambda) = \mu_1.$$

Since the filtration is Brownian, enlarged only by the initial variables, the martingale representation theorem gives a continuous version of X^λ . Hence $X^\lambda \in \mathcal{X}(\mu_0, \mu_1; \kappa)$.

Using (6),

$$\mathbb{E}\langle X^\lambda, Y \rangle_1 = \mathbb{E}[X_1^\lambda(Y_1 - Y_0)] = \iint \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a)} \int yz q(dy, dz) \kappa(dx, da).$$

Since X^λ is admissible,

$$V_{MBB}^\kappa(\mu_0, \mu_1) \geq \mathbb{E}\langle X^\lambda, Y \rangle_1.$$

Thus, for every $\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)$,

$$V_{MBB}^\kappa(\mu_0, \mu_1) \geq \iint \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a)} \int yz q(dy, dz) \kappa(dx, da).$$

Taking the supremum over $\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)$, we obtain

$$V_{MBB}^\kappa(\mu_0, \mu_1) \geq V_{WOT}^\kappa(\mu_0, \mu_1).$$

Combining the two inequalities gives the result. $\square$

Remark 3.6. In the usual WOT problem, a coupling $\pi \in \text{Cpl}(\mu, \nu)$ is written as $\pi(dx, dy) = \mu(dx) \pi_x(dy)$. Equivalently, one may view this as a Monge-type coupling in $\mathcal{P}(X \times \mathcal{P}(Y))$, supported on the graph of the map $x \mapsto \pi_x$. In the present setting, the conditional law of the terminal value is not determined only by the initial position x . After desintegration $\kappa(dx, da) = \mu_0(dx) \kappa_x(da)$, for each fixed x we obtain a random conditional law $a \mapsto \lambda_{x,a}, a \sim \kappa_x$. Hence, after projecting back to the variable x , the kernel is no longer a single Monge-type choice $x \mapsto \pi_x$, but rather a Kantorovich coupling on $\mathcal{P}(X \times \mathcal{P}(Y))$.

3.4 Existence of a static optimizer

We now prove that the static problem admits an optimizer. The proof relies on the lower semicontinuity theorem for weak optimal transport from [11]. The only point specific to our setting is the dependence of the reference kernel θ_a on the initial volatility state a .

Lemma 3.7. *Assume that $b, \eta : \mathbb{R} \rightarrow \mathbb{R}$ are globally Lipschitz and have at most linear growth. For each $a \in \mathbb{R}$, let ξ^a be the unique strong solution of*

$$d\xi_t^a = b(\xi_t^a) dt + \eta(\xi_t^a) dW_t, \quad \xi_0^a = a, \quad t \in [0, 1].$$

Define

$$Z^a = \int_0^1 \xi_t^a dB_t, \quad \theta_a = \text{Law}(Z^a).$$

Then the map

$$a \rightarrow \theta_a$$

is continuous from $\mathbb{R}$ into $(\mathcal{P}_2(\mathbb{R}), \mathcal{W}_2)$.

Proof. Let $a, a' \in \mathbb{R}$. By the standard stability estimate for strong solutions of SDEs with globally Lipschitz coefficients, there exists a constant $C > 0$, independent of a, a' , such that

$$\sup_{t \in [0, 1]} \mathbb{E} \left[|\xi_t^a - \xi_t^{a'}|^2 \right] \leq C|a - a'|^2.$$

By Itô's isometry,

$$\mathbb{E} \left[|Z^a - Z^{a'}|^2 \right] = \mathbb{E} \left[\int_0^1 |\xi_t^a - \xi_t^{a'}|^2 dt \right] \leq C|a - a'|^2.$$

Since $(Z^a, Z^{a'})$ is a coupling of θ_a and $\theta_{a'}$, we obtain

$$\mathcal{W}_2^2(\theta_a, \theta_{a'}) \leq \mathbb{E} \left[|Z^a - Z^{a'}|^2 \right] \leq C|a - a'|^2.$$

Hence $a \mapsto \theta_a$ is continuous in $\mathcal{W}_2$. $\square$

Corollary 3.8. *The map*

$$(\rho, a) \rightarrow \mathcal{W}_2^2(\rho, \theta_a)$$

is continuous on $\mathcal{P}_2(\mathbb{R}) \times \mathbb{R}$. Moreover, for each fixed $a \in \mathbb{R}$, the map

$$\rho \rightarrow \mathcal{W}_2^2(\rho, \theta_a)$$

is convex on $\mathcal{P}_2(\mathbb{R})$.

Proof. Let $\rho_n \rightarrow \rho$ in $\mathcal{P}_2(\mathbb{R})$ and let $a_n \rightarrow a$. By Lemma 3.7,

$$\mathcal{W}_2(\theta_{a_n}, \theta_a) \rightarrow 0.$$

Therefore,

$$|\mathcal{W}_2(\rho_n, \theta_{a_n}) - \mathcal{W}_2(\rho, \theta_a)| \leq \mathcal{W}_2(\rho_n, \rho) + \mathcal{W}_2(\theta_{a_n}, \theta_a) \rightarrow 0.$$

Hence

$$\mathcal{W}_2^2(\rho_n, \theta_{a_n}) \rightarrow \mathcal{W}_2^2(\rho, \theta_a).$$

For convexity, let $\rho_0, \rho_1 \in \mathcal{P}_2(\mathbb{R})$ and $t \in [0, 1]$. Let π_0 and π_1 be optimal couplings between ρ_0 and θ_a , and between ρ_1 and θ_a , respectively. Then

$$\pi = t\pi_0 + (1-t)\pi_1$$

is a coupling between $t\rho_0 + (1-t)\rho_1$ and θ_a . Hence

$$\begin{aligned} \mathcal{W}_2^2(t\rho_0 + (1-t)\rho_1, \theta_a) &\leq \int |x - y|^2 \pi(dx, dy) \\ &= t \int |x - y|^2 \pi_0(dx, dy) + (1-t) \int |x - y|^2 \pi_1(dx, dy) \\ &= t\mathcal{W}_2^2(\rho_0, \theta_a) + (1-t)\mathcal{W}_2^2(\rho_1, \theta_a). \end{aligned}$$

Thus $\rho \mapsto \mathcal{W}_2^2(\rho, \theta_a)$ is convex. □

Corollary 3.9. *The cost*

$$C((x, a), \rho) = \mathcal{W}_2^2(\rho, \theta_a), \quad (x, a) \in \mathbb{R}^2, \quad \rho \in \mathcal{P}_2(\mathbb{R}),$$

is non-negative, lower semicontinuous on $\mathbb{R}^2 \times \mathcal{P}_2(\mathbb{R})$, and convex in the variable ρ .

Proof. The cost does not depend on x . The result follows directly from Corollary 3.8. □

We can now prove existence of an optimizer for the static problem.

Theorem 3.10 (Existence of a static optimizer). *Assume that b and η are globally Lipschitz and have at most linear growth. Assume also that $\kappa \in \mathcal{P}_2(\mathbb{R}^2)$, that the first marginal of κ is μ_0 , that $\mu_1 \in \mathcal{P}_2(\mathbb{R})$, and that*

$$\text{Cpl}_{M_1}(\kappa, \mu_1) \neq \emptyset.$$

Then the minimization problem

$$\inf_{\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)} \iint \mathcal{W}_2^2(\lambda_{x,a}, \theta_a) \kappa(dx, da)$$

admits an optimizer.

Proof. By Corollary 3.9, the cost

$$C((x, a), \rho) = \mathcal{W}_2^2(\rho, \theta_a)$$

is non-negative, lower semicontinuous, and convex in the measure variable. Therefore, by Theorem 3.1, the functional

$$\lambda \longmapsto \iint \mathcal{W}_2^2(\lambda_{x,a}, \theta_a) \kappa(dx, da)$$

is lower semicontinuous under convergence in $\mathcal{P}_2(\mathbb{R}^3)$ among probability measures with fixed (x, a) -marginal κ .

We next show that the admissible set is compact. Since the first marginal κ and the third marginal μ_1 are fixed, the set $\text{Cpl}(\kappa, \mu_1)$ is tight. Moreover, for every $\lambda \in \text{Cpl}(\kappa, \mu_1)$,

$$\int_{\mathbb{R}^3} (x^2 + a^2 + y^2) \lambda(dx, da, dy) = \int_{\mathbb{R}^2} (x^2 + a^2) \kappa(dx, da) + \int_{\mathbb{R}} y^2 \mu_1(dy).$$

Thus the second moments are fixed, and $\text{Cpl}(\kappa, \mu_1)$ is compact in $\mathcal{P}_2(\mathbb{R}^3)$.

It remains to check that the martingale constraint is closed. Let $\lambda^n \in \text{Cpl}_{M_1}(\kappa, \mu_1)$ and suppose that

$$\lambda^n \rightarrow \lambda \quad \text{in } \mathcal{P}_2(\mathbb{R}^3).$$

Since the marginals are preserved under weak convergence, we have $\lambda \in \text{Cpl}(\kappa, \mu_1)$. Moreover, for every $\varphi \in C_b(\mathbb{R}^2)$,

$$\int \varphi(x, a)(y - x) \lambda^n(dx, da, dy) = 0.$$

Passing to the limit gives

$$\int \varphi(x, a)(y - x) \lambda(dx, da, dy) = 0.$$

Hence

$$\int y \lambda_{x,a}(dy) = x, \quad \kappa\text{-a.e.}$$

Therefore $\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)$, and $\text{Cpl}_{M_1}(\kappa, \mu_1)$ is closed in $\mathcal{P}_2(\mathbb{R}^3)$. Consequently, $\text{Cpl}_{M_1}(\kappa, \mu_1)$ is compact.

Let (λ^n) be a minimizing sequence. By compactness, there exists a subsequence (λ^{n_k}) , convergent to some $\lambda^* \in \text{Cpl}_{M_1}(\kappa, \mu_1)$ in $\mathcal{P}_2(\mathbb{R}^3)$. By lower semicontinuity,

$$\iint \mathcal{W}_2^2(\lambda_{x,a}^*, \theta_a) \kappa(dx, da) \leq \liminf_{n \rightarrow \infty} \iint \mathcal{W}_2^2(\lambda_{x,a}^{n_k}, \theta_a) \kappa(dx, da).$$

Thus λ^* attains the infimum. $\square$

4 Bass-type optimizers

In the present section, we identify a class of optimizers for the generalized martingale Benamou–Brenier problem that have the same structure as a standard stretched Brownian motion in the classical martingale Benamou–Brenier problem.

We prove a verification result: whenever the map r can be chosen so that the corresponding Bass-type martingale has the prescribed initial condition, then this martingale is optimal. The existence and construction of such maps will be discussed in the numerical section.

4.1 The general Bass-type verification theorem

Let $\kappa = \text{Law}(X_0, \xi_0)$ be the joint initial law of the price and volatility. For $a > 0$, denote by $\xi^a = (\xi_s^a)_{0 \leq s \leq 1}$ the volatility process started from $\xi_0 = a$, and set

$$\theta_a^u = (u + \text{id})_{\#} \theta_a \quad \text{where} \quad \theta_a = \text{Law}\left(\int_0^1 \xi_s^a dB_s\right),$$

Theorem 4.1 (Bass-type optimizer with random initial volatility). *Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$ satisfy $\mu_0 \preceq_{cx} \mu_1$, and let κ have first marginal μ_0 . Define the reference process*

$$Y_t = r(X_0, \xi_0) + \int_0^t \xi_s dB_s, \quad 0 \leq t \leq 1.$$

Set

$$\nu = \text{Law}(Y_1), \quad g_0 = F_{\mu_1}^{-1} \circ F_{\nu}.$$

Suppose that r is chosen so that

$$\mathbb{E} \left[g_0 \left(r(x, a) + \int_0^1 \xi_s^a dB_s \right) \right] = x, \quad \kappa\text{-a.e. } (x, a). \quad (7)$$

Define

$$M_t = \mathbb{E}[g_0(Y_1) \mid \mathcal{F}_t], \quad 0 \leq t \leq 1.$$

Then

$$M_0 = X_0, \quad \text{Law}(M_0, \xi_0) = \kappa, \quad \text{Law}(M_1) = \mu_1.$$

Moreover, M is an optimizer of the generalized martingale Benamou–Brenier problem.

Remark 4.2. The theorem should be read as a verification result: if such a map r exists, then the associated Bass-type martingale is optimal.

Proof. Conditionally on $(X_0, \xi_0) = (x, a)$, we have

$$Y_0 = r(x, a), \quad Y_1 = r(x, a) + Z^a, \quad \text{Law}(Y_1 \mid X_0 = x, \xi_0 = a) = \theta_a^{r(x, a)}.$$

Starting from the static formulation (SV-WOT) and shifting the second marginal by $r(x, a)$, we obtain

$$\begin{aligned} V_{WOT}^{\kappa}(\mu_0, \mu_1) &= \sup_{\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)} \int \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a^{r(x,a)})} \int y(z - r(x, a)) q(dy, dz) \kappa(dx, da) \\ &= \sup_{\lambda \in \text{Cpl}_{M_1}(\kappa, \mu_1)} \int \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a^{r(x,a)})} \int yz q(dy, dz) \kappa(dx, da) - \int xr(x, a) \kappa(dx, da), \end{aligned} \quad (8)$$

where we used the martingale constraint

$$\int y \lambda_{x,a}(dy) = x, \quad \kappa\text{-a.e. } (x, a).$$

Let ψ and ψ^* be conjugate convex functions. Fenchel's inequality,

$$yz \leq \psi(z) + \psi^*(y),$$

implies that, for every admissible λ ,

$$\int \sup_{q \in \text{Cpl}(\lambda_{x,a}, \theta_a^{r(x,a)})} \int yz q(dy, dz) \kappa(dx, da) \leq \int \psi^*(y) \mu_1(dy) + \iint \psi(z) \theta_a^{r(x,a)}(dz) \kappa(dx, da).$$

Together with (8), this gives

$$V_{WOT}^\kappa(\mu_0, \mu_1) \leq \int \psi^*(y) \mu_1(dy) + \iint \psi(z) \theta_a^{r(x,a)}(dz) \kappa(dx, da) - \int xr(x, a) \kappa(dx, da). \quad (9)$$

We now show that the martingale from the statement attains this bound. Since

$$g_0 = F_{\mu_1}^{-1} \circ F_\nu,$$

we have $(g_0)_\# \nu = \mu_1$. Hence

$$M_1 = g_0(Y_1), \quad \text{Law}(M_1) = \mu_1.$$

Moreover, by (7),

$$M_0 = \mathbb{E}[g_0(Y_1) \mid X_0, \xi_0] = \mathbb{E}[g_0(r(X_0, \xi_0) + Z^{\xi_0}) \mid X_0, \xi_0] = X_0.$$

Thus $\text{Law}(M_0, \xi_0) = \kappa$, so M is admissible.

Since g_0 is increasing, choose a convex function ψ such that

$$g_0 = \nabla \psi,$$

and let ψ^* be its convex conjugate. Define

$$\lambda := \text{Law}(M_0, \xi_0, M_1), \quad \lambda(dx, da, dy) = \kappa(dx, da) \lambda_{x,a}(dy).$$

For κ -a.e. (x, a) , set

$$q_{x,a} := \text{Law}(M_1, Y_1 \mid M_0 = x, \xi_0 = a) = \text{Law}(g_0(r(x, a) + Z^a), r(x, a) + Z^a).$$

Then

$$q_{x,a} \in \text{Cpl}(\lambda_{x,a}, \theta_a^{r(x,a)}).$$

We now plug this coupling into the right-hand side of (9). Since $q_{x,a}$ has marginals $\lambda_{x,a}$ and $\theta_a^{r(x,a)}$, and since the second marginal of λ is μ_1 , we get

$$\begin{aligned} & \int \psi^*(y) \mu_1(dy) + \iint \psi(z) \theta_a^{r(x,a)}(dz) \kappa(dx, da) - \int xr(x, a) \kappa(dx, da) \\ &= \int \iint (\psi^*(y) + \psi(z)) q_{x,a}(dy, dz) \kappa(dx, da) - \int xr(x, a) \kappa(dx, da). \end{aligned}$$

Since $M_1 = g_0(Y_1) = \nabla \psi(Y_1)$, the coupling $q_{x,a}$ is supported on the graph $y = \nabla \psi(z)$. Hence Fenchel equality holds:

$$yz = \psi(z) + \psi^*(y), \quad q_{x,a}\text{-a.s.}$$

Consequently,

$$\begin{aligned} & \int \psi^*(y) \mu_1(dy) + \iint \psi(z) \theta_a^{r(x,a)}(dz) \kappa(dx, da) - \int xr(x, a) \kappa(dx, da) \\ &= \int \iint yz q_{x,a}(dy, dz) \kappa(dx, da) - \int xr(x, a) \kappa(dx, da). \end{aligned} \quad (10)$$

On the other hand, the family $q_{x,a}$ is admissible in the inner supremum in (8). Thus the right-hand side of (10) is also a lower bound for $V_{WOT}^\kappa(\mu_0, \mu_1)$. Together with (9), this proves that the upper bound is attained by the coupling supported on the graph of the transport map g_0 .

Therefore,

$$\begin{aligned} V_{WOT}^\kappa(\mu_0, \mu_1) &= \int \iint yz q_{x,a}(dy, dz) \kappa(dx, da) - \int xr(x, a) \kappa(dx, da) \\ &= \int \iint y(z - r(x, a)) q_{x,a}(dy, dz) \kappa(dx, da), \end{aligned}$$

where the second equality follows from $\int y \lambda_{x,a}(dy) = x$ and from the fact that the first marginal of $q_{x,a}$ is $\lambda_{x,a}$.

Hence

$$V_{WOT}^\kappa(\mu_0, \mu_1) = \mathbb{E}[M_1(Y_1 - Y_0)].$$

Since $M_0 = X_0$ and $Y_1 - Y_0$ has conditional mean zero given (X_0, ξ_0) ,

$$\mathbb{E}[M_0(Y_1 - Y_0)] = 0.$$

Consequently,

$$\mathbb{E}[M_1(Y_1 - Y_0)] = \mathbb{E}[(M_1 - M_0)(Y_1 - Y_0)] = \mathbb{E}\langle M, Y \rangle_1.$$

Therefore M attains the value of the generalized martingale Benamou–Brenier problem, and is an optimizer. $\square$

4.2 Constant initial volatility

We now record the deterministic initial-volatility case as a direct consequence of Theorem 4.1. In this case

$$\xi_0 = c > 0, \quad \kappa = \mu_0 \otimes \delta_c.$$

The conditional increment law is independent of the initial state. Writing

$$Z := \int_0^1 \xi_s dB_s, \quad \theta := \text{Law}(Z),$$

we have $\theta_a = \theta$ for $a = c$. Hence the static formulation (SV-WOT) reduces to

$$V_{WOT}^\kappa(\mu_0, \mu_1) = \sup_{\pi \in \text{Cpl}_M(\mu_0, \mu_1)} \int \sup_{q \in \text{Cpl}(\pi_x, \theta)} \int yz q(dy, dz) \mu_0(dx).$$

This is exactly the static formulation of the original martingale Benamou–Brenier problem, with the Gaussian increment law γ replaced by the reference increment law θ .

Corollary 4.3 (Constant initial volatility). *Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R})$ satisfy $\mu_0 \preceq_{cx} \mu_1$. Assume that $\xi_0 = c > 0$ is deterministic and let*

$$Y_t = r(X_0) + \int_0^t \xi_s dB_s, \quad 0 \leq t \leq 1,$$

where $X_0 \sim \mu_0$. Set

$$\nu = \text{Law}(Y_1), \quad g_0 = F_{\mu_1}^{-1} \circ F_\nu.$$

Suppose that r is chosen so that

$$\mathbb{E} \left[g_0 \left(r(x) + \int_0^1 \xi_s dB_s \right) \right] = x, \quad \mu_0\text{-a.e. } x.$$

Define

$$M_t = \mathbb{E}[g_0(Y_1) \mid \mathcal{F}_t], \quad 0 \leq t \leq 1.$$

Then

$$M_0 = X_0, \quad \text{Law}(M_0) = \mu_0, \quad \text{Law}(M_1) = \mu_1.$$

Moreover, M is an optimizer of the generalized martingale Benamou–Brenier problem.

Proof. This is Theorem 4.1 applied to $\kappa = \mu_0 \otimes \delta_c$. In this case $r(x, a) = r(x)$, the conditional increment law θ_a is the fixed law θ , and condition (7) reduces to the condition above. $\square$

In the next section we study the existence and construction of such maps r . We also explain how the construction can be performed on several time intervals in order to obtain a continuous-time price process.

5 Practical calibration of the model

We describe a practical calibration procedure for the multi-marginal stochastic volatility Bass construction. Given maturities

$$0 = t_0 < t_1 < \cdots < t_n$$

and market-implied marginals

$$\mu_{t_0} \preceq_{cx} \mu_{t_1} \preceq_{cx} \cdots \preceq_{cx} \mu_{t_n},$$

the objective is to construct a continuous martingale

$$S = (S_t)_{0 \leq t \leq t_n}$$

such that

$$S_{t_i} \sim \mu_{t_i}, \quad i = 0, \dots, n,$$

while preserving realistic stochastic volatility dynamics between maturities.

The calibration proceeds in two stages. First, a stochastic volatility reference model is calibrated to market data and determines the conditional increment laws. Second, on each interval, a Bass-type correction is computed to match the prescribed marginals exactly.

Calibration principle We begin with a stochastic volatility reference model

$$\begin{aligned} dY_t &= \xi_t dB_t, \\ d\xi_t &= b(\xi_t) dt + \eta(\xi_t) dW_t, \quad dB_t dW_t = \rho dt. \end{aligned}$$

For every interval $[t_i, t_{i+1}]$ and initial volatility state a , let

$$Z_i^a = \int_{t_i}^{t_{i+1}} \xi_s^{i,a} dB_s, \quad \theta_i^a = \text{Law}(Z_i^a),$$

where $\xi^{i,a}$ denotes the volatility process started from $\xi_{t_i} = a$. The conditional increment laws θ_i^a serve as reference kernels for the calibration.

We seek a correction map $r_i(\cdot, a) : \mathbb{R} \rightarrow \mathbb{R}$ such that, for $X \sim \mu_i$,

$$\nu_i^a = \text{Law}(r_i(X, a) + Z_i^a), \quad g_i^a = F_{\mu_{i+1}}^{-1} \circ F_{\nu_i^a},$$

satisfy the martingale condition

$$\mathbb{E}[g_i^a(r_i(x, a) + Z_i^a)] = x, \quad \mu_i\text{-a.e. } x.$$

The calibrated process on $[t_i, t_{i+1}]$ is then

$$S_t = \mathbb{E}[g_i^a(r_i(S_{t_i}, a) + Z_i^a) \mid \mathcal{F}_t].$$

Proposition 5.1. *The law $\alpha_i^a = \text{Law}(r_i(X, a))$ can be computed through the fixed-point operator*

$$(\mathcal{A}_i^a F)(x) = F_{\mu_i} \circ K_i^{+,a} \left(F_{\mu_{i+1}}^{-1} \circ (K_i^{-,a} F) \right)(x),$$

where

$$(K_i^{\pm,a} \psi)(x) = \mathbb{E}[\psi(x \pm Z_i^a)].$$

If $F_{\alpha_i^a}$ is the fixed point, then

$$r_i(x, a) = F_{\alpha_i^a}^{-1} \circ F_{\mu_i}(x).$$

We show the construction of the operator similarly to [17].

Proof. Let

$$F(x) = \mathbb{P}(r_i(X, a) < x).$$

We first compute the map g_i^a . By definition,

$$\mathbb{P}(g_i^a(r_i(X, a) + Z_i^a) < x) = F_{\mu_{i+1}}(x).$$

Since g_i^a is increasing, this is equivalent to

$$\mathbb{P}(r_i(X, a) + Z_i^a < (g_i^a)^{-1}(x)) = F_{\mu_{i+1}}(x).$$

The density of Z_i^a is denoted by ψ_i^a . Rewriting the probability as an integral, we get

$$\int_{\mathbb{R}} F((g_i^a)^{-1}(x) - z) \psi_i^a(z) dz = F_{\mu_{i+1}}(x).$$

Hence

$$(K_i^{-,a}F)((g_i^a)^{-1}(x)) = F_{\mu_{i+1}}(x), \quad (K_i^{-,a}F)(y) = \int_{\mathbb{R}} F(y-z)\psi_i^a(z) dz.$$

Thus

$$g_i^a = F_{\mu_{i+1}}^{-1} \circ (K_i^{-,a}F).$$

To match the first marginal,

$$F_{\mu_i}(x) = \mathbb{P}(\mathbb{E}[g_i^a(r_i(X, a) + Z_i^a)] < x).$$

Denote

$$h(c) = \mathbb{E}[g_i^a(c + Z_i^a)] = \int_{\mathbb{R}} g_i^a(c + z)\psi_i^a(z) dz = K_i^{+,a}[F_{\mu_{i+1}}^{-1} \circ (K_i^{-,a}F)](c).$$

Therefore

$$F_{\mu_i}(x) = \mathbb{P}(h(r_i(X, a)) < x).$$

Since h is increasing, we get

$$F_{\mu_i}(h(x)) = \mathbb{P}(h(r_i(X, a)) < h(x)) = \mathbb{P}(r_i(X, a) < x) = F(x).$$

Hence

$$F(x) = F_{\mu_i} \circ K_i^{+,a}(F_{\mu_{i+1}}^{-1} \circ (K_i^{-,a}F))(x).$$

Finally, since $r_i(\cdot, a)$ transports μ_i to α_i^a , we obtain

$$r_i(x, a) = F_{\alpha_i^a}^{-1} \circ F_{\mu_i}(x).$$

□

Numerical implementation For fixed volatility state a , the increment law θ_i^a is translation invariant and admits a density ψ_i^a . The operator $\mathcal{A}_i^a$ can therefore be approximated by discrete convolutions following [47]. On a grid with spacing h ,

$$(K_i^{+,a}F)(w) \approx \sum_{n=-N}^N F(w + nh)\psi_i^a(nh)h.$$

Substituting this approximation into the fixed-point operator yields the iterative scheme

$$F^{(q+1)} = \mathcal{A}_i^a F^{(q)},$$

which is repeated until convergence. After calibration, the process is simulated through its Markovian representation

$$S_t = f_i(t, Y'_t, \xi_t),$$

where

$$Y'_t = Y'_{t_i} + \int_{t_i}^t \xi_s dB_s, \quad Y'_{t_i} \sim \alpha_i^a,$$

and

$$f_i(t, y, a) = \mathbb{E}[g_i^a(Y'_{t_{i+1}}) \mid Y'_t = y, \xi_t = a].$$

Multi-maturity construction On the first interval $[0, t_1]$, the initial volatility is deterministic: $\xi_0 = c$, so only one fixed-point problem is solved. At later maturities, the correction depends on the realized volatility state:

$$r_i = r_i(x, a).$$

To make the calibration finite-dimensional, we discretize the volatility distribution at switching times:

$$\beta_i^{(m)} = \sum_{k=1}^m p_k \delta_{a_k}, \quad \sum_{k=1}^m p_k = 1.$$

For each interval and volatility level a_k , we compute

$$\left(\theta_i^{a_k}, \alpha_i^{a_k}, r_i(\cdot, a_k), g_i^{a_k} \right).$$

These one-step calibration problems are independent and can therefore be solved offline and in parallel. Suppose the process reaches

$$(S_{t_i}^-, \xi_{t_i}^-)$$

at time t_i . We select the nearest volatility bin

$$k^* = \arg \min_k |\xi_{t_i}^- - a_k|.$$

Since the calibration on the next interval is parametrized by the variable $Y'_{t_i} \sim \alpha_i^{a_k}$, it should be chosen such that the process S_t is continuous across intervals. Define the map

$$s_{i,k}(y) = \mathbb{E}[g_i^{a_k}(y + Z_i^{a_k})].$$

The initial state of the reference process Y_{t_i} is then recovered through

$$Y_{t_i} = s_{i,k^*}^{-1}(S_{t_i}^-).$$

The continuation on $[t_i, t_{i+1}]$ is

$$S_t = \mathbb{E}\left[g_i^{a_{k^*}}(Y_{t_i} + Z_i^{a_{k^*}}) \mid \mathcal{F}_t\right].$$

By construction,

$$S_{t_i}^+ = s_{i,k^*}(Y_{t_i}) = S_{t_i}^-,$$

so the price process is continuous at every switching time.

In practice, the maps $s_{i,k}$, their inverses, and the functions f_i are stored during the offline calibration stage. Simulation then consists only of selecting the volatility bin, evaluating a one-dimensional inverse map, and propagating the corresponding conditional expectation process.

6 Numerical examples

We illustrate the calibration procedure on synthetic market data. In all experiments, the reference dynamics are given by an arithmetic SABR model, while the target marginals are generated by different market specifications. We consider three examples: calibration to another SABR model on a single interval, calibration to two Heston marginals [35], and calibration to marginals implied by SVI volatility smiles.

Reference stochastic volatility model The reference dynamics are given by the arithmetic SABR model [29]

$$\begin{aligned} dY_t &= \xi_t dW_t, \\ d\xi_t &= \nu \xi_t dB_t, \quad dW_t dB_t = \rho dt. \end{aligned}$$

The parameters of the reference model are fixed:

$$\xi_0 = 0.2, \quad \rho = -0.4, \quad \nu = 0.5.$$

The SABR model is chosen as a reference because it admits a closed-form approximation of implied volatility introduced by Hagan et al. [29], allowing one to recover both marginal laws and transition kernels required by the calibration.

Theorem 6.1 (Hagan’s implied volatility approximation for $\beta = 0$ [29]). *For maturity T and strike K , the SABR implied volatility with $\beta = 0$ is approximated by the lognormal (Black) implied volatility*

$$\begin{aligned} \sigma^{iv}(T, K) &= \frac{\alpha}{\sqrt{fK}} \left(1 + \frac{1}{24} \log^2 \left(\frac{f}{K} \right) + \frac{1}{1920} \log^4 \left(\frac{f}{K} \right) + \dots \right) \frac{z}{x(z)} \\ &\times \left(1 + \left[\frac{1}{24} \frac{\alpha^2}{fK} + \frac{2-3\rho^2}{24} \nu^2 \right] T + \dots \right), \end{aligned}$$

where

$$z = \frac{\nu}{\alpha} \sqrt{fK} \log \left(\frac{f}{K} \right), \quad x(z) = \log \left(\frac{\sqrt{1-2\rho z + z^2} + z - \rho}{1 - \rho} \right).$$

Equivalently, this can be written in the numerically stable form

$$\sigma^{iv}(T, K) = \alpha \frac{\log(f/K) \cdot z}{(f-K) \cdot x(z)} \left[1 + T \left(\frac{\alpha^2}{24fK} + \frac{2-3\rho^2}{24} \nu^2 \right) \right],$$

where the factor $\frac{\log(f/K) \cdot z}{(f-K) \cdot x(z)}$ approaches $\frac{1}{f}$ in the limit $K \rightarrow f$.

For each maturity T , Hagan’s formula provides an implied volatility smile $\sigma_{\text{SABR}}^{iv}(T, K)$. Substituting this volatility into the Black-Scholes pricing formula gives

$$C^{\text{SABR}}(T, K) = C_{\text{BS}}(T, K, Y_0, \sigma_{\text{SABR}}^{iv}(T, K)).$$

The cumulative distribution function is then recovered through the Breeden–Litzenberger relation:

$$F_{Y_T}(K) = 1 + \partial_K C^{\text{SABR}}(T, K),$$

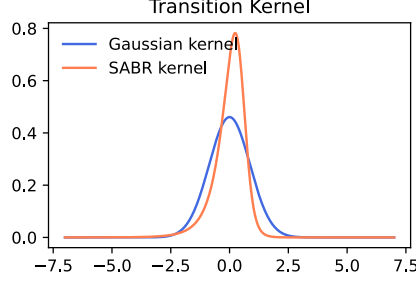
These numerical distribution and quantile functions are used to compute the transport map:

$$g = F_{\mu_T}^{-1} \circ F_{Y_T}.$$

For later intervals, the same procedure is used to recover the conditional increment law:

$$\theta_i^a = \text{Law} \left(\int_{t_i}^{t_{i+1}} \xi_s^{i,a} dW_s \right).$$

In the numerical implementation we use one volatility bin and take $a_i = \mathbb{E}[\xi_{t_i}]$. The corresponding density serves as the kernel in the fixed-point iteration.



Simulation procedure We simulate the reference dynamics using an Euler–Maruyama scheme:

$$Y_t = Y_0 + \rho \int_0^t \xi_s dB_s + \sqrt{1 - \rho^2} \int_0^t \xi_s dW_s, \quad \xi_t = \xi_0 \exp \left(\nu B_t - \frac{1}{2} \nu^2 t \right).$$

Where B and Z are two independent brownian motions. For $\Delta t = \frac{T}{N_{\text{steps}}}$, the discretization is

$$\begin{aligned} \xi_n &= \xi_{n-1} \exp \left(\nu Z_1^{(n-1)} \sqrt{\Delta t} - \frac{1}{2} \nu^2 \Delta t \right), \\ Y_n &= Y_{n-1} + \xi_{n-1} \left(\rho Z_1^{(n-1)} + \sqrt{1 - \rho^2} Z_2^{(n-1)} \right) \sqrt{\Delta t}. \end{aligned}$$

Numerical setup Unless stated otherwise, we use $N_{\text{steps}} = 100$, $N_{\text{paths}} = 10^5$. The fixed-point iteration uses $N = 10^4$, $h = 7 \cdot 10^{-4}$, with stopping criterion $\|F^n - F^{n-1}\|_{L^\infty} < 10^{-3}$.

6.1 Example 1: calibration to SABR model

The first experiment illustrates the dynamical properties of the construction. The target marginal is generated by another arithmetic SABR model with parameters

$$\xi_0^{\text{mkt}} = 0.2, \quad \rho^{\text{mkt}} = -0.2, \quad \nu^{\text{mkt}} = 0.55.$$

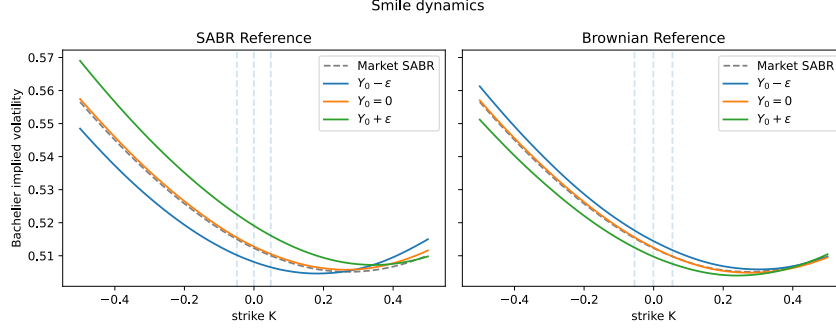
We consider one maturity $t_1 = 1$ and define $\mu_{t_1} = \text{Law}(S_{t_1}^{\text{mkt}})$. The calibrated process is

$$S_t = \mathbb{E} \left[g(Y_{t_1}) \mid \mathcal{F}_t \right], \quad g = F_{\mu_{t_1}}^{-1} \circ F_{Y_{t_1}}.$$

Using the Markovian representation $S_t = f(t, Y_t, \xi_t)$, We simulate the Ito representation:

$$dS_t = \partial_y f(t, Y_t, \xi_t) \xi_t dW_t + \partial_a f(t, Y_t, \xi_t) \nu \xi_t dB_t.$$

We simulate this SDE directly to study the induced stochastic volatility dynamics.



6.2 Example 2: calibration to Heston marginals

We next calibrate the model to two marginals generated by the Heston model [35]

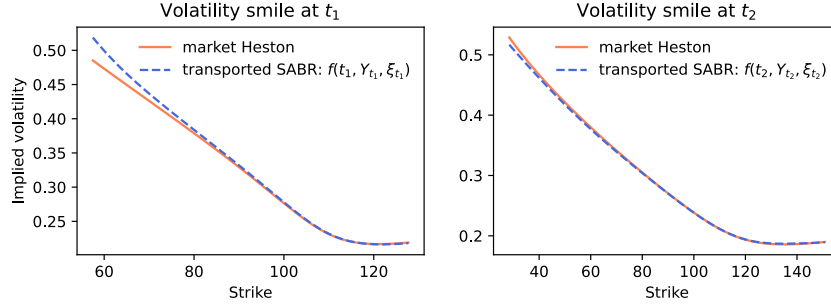
$$\begin{aligned} dS_t &= \sqrt{v_t} S_t dW_t^{(1)}, \\ dv_t &= \kappa(\theta - v_t)dt + \eta\sqrt{v_t}dW_t^{(2)}, \\ dW_t^{(1)}dW_t^{(2)} &= \rho dt. \end{aligned}$$

We use $S_0 = 100$, $\kappa = 2$, $\theta = 0.07$, $\rho = -0.7$, $\eta = 1$, $v_0 = 0.1$. The maturities are $t_1 = 0.25$, $t_2 = 1$. The target marginals are

$$\mu_{t_1} = \text{Law}(S_{t_1}), \quad \mu_{t_2} = \text{Law}(S_{t_2}),$$

recovered using the Breeden–Litzenberger formula.

On $[0, t_1]$, we calibrate from δ_{S_0} to μ_{t_1} . On $[t_1, t_2]$, we calibrate from μ_{t_1} to μ_{t_2} using one volatility bin $a_1 = E[\xi_{t_1}]$. In this example we only test marginal calibration and do not simulate the Itô representation.



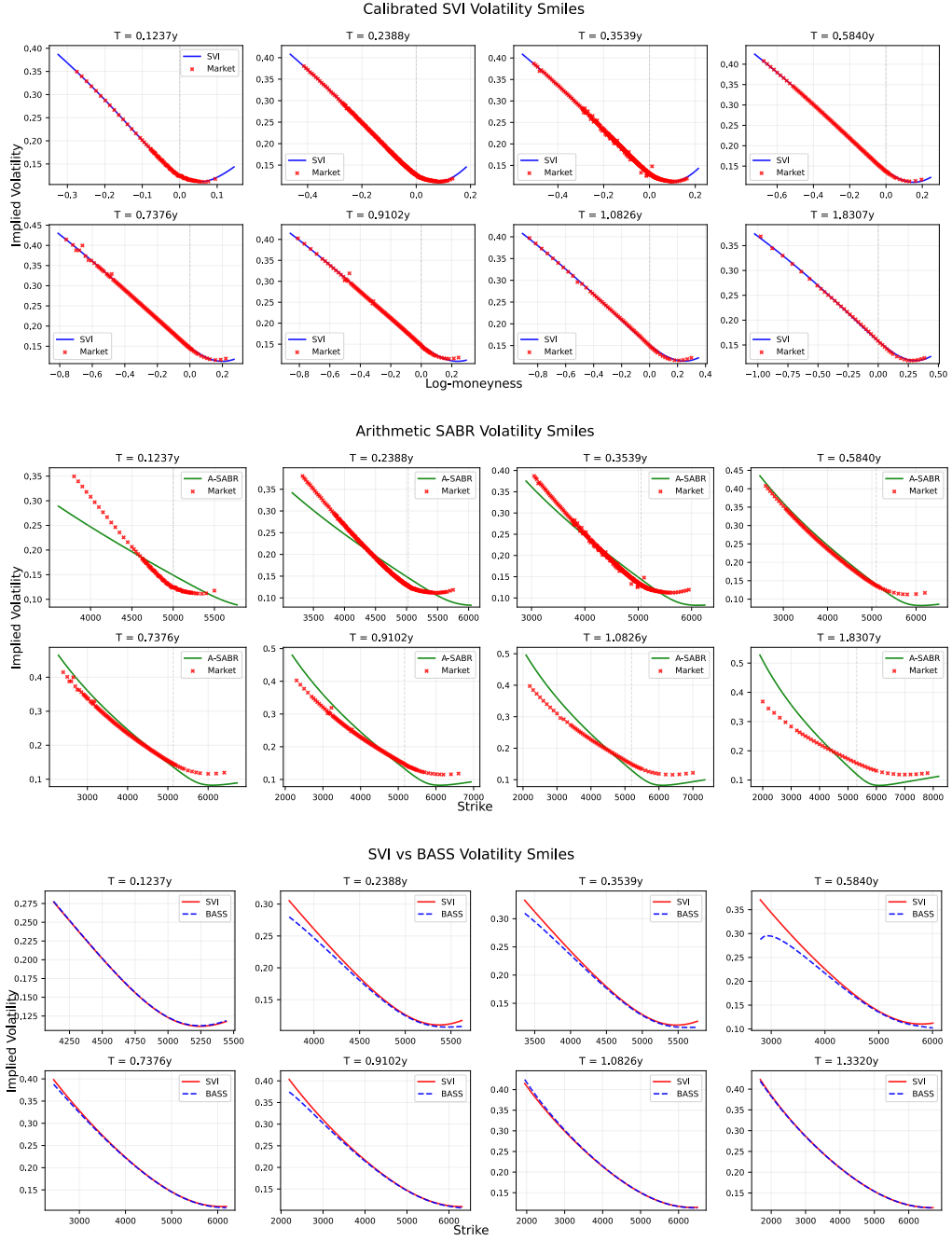
6.3 Example 3: calibration to SVI

In this example we calibrate the model to S&P 500 market smiles as of 21.02.2024. We first fit the observed implied volatilities by the SVI parametrization in order to obtain a smooth arbitrage-free approximation of the market smile at each maturity.

The reference SABR model is first calibrated to the S&P 500 implied volatility surface obtaining the parameres:

$$\xi_0 = 706.6, \quad \rho = -0.706 \quad \nu = 1.0482$$

Then the calibration of the Bass model is done for 8 maturities using one volatility bin $a_1 = E[\xi_{t_1}]$.



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